

1.apt-get update

This command is used to resynchronize the package index files from their sources. It updates the list of available packages and their versions from the defined repositories.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:3 https://brave-browser-apt-release.s3.brave.com stable InRelease
Get:4 https://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Hit:5 https://packages.microsoft.com/repos/edge stable InRelease
Get:6 https://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,215 B]
Hit:7 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Get:8 https://packages.microsoft.com/repos/code stable/main arm64 Packages [24.1 kB]
Get:9 https://packages.microsoft.com/repos/code stable/main armhf Packages [24.2 kB]
Get:10 https://packages.microsoft.com/repos/code stable/main amd64 Packages [24.0 kB]
Ign:11 https://zotero.retorque.re/file/apt-package-archive ./ InRelease
Hit:12 https://packages.cloud.google.com/apt cloud-sdk InRelease
Hit:13 https://ppa.launchpadcontent.net/git-core/ppa/ubuntu noble InRelease
Get:15 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:16 https://ppa.launchpadcontent.net/hluk/copyq/ubuntu noble InRelease
Get:17 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:18 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:19 https://ppa.launchpadcontent.net/touchegg/stable/ubuntu noble InRelease
Hit:14 https://packagecloud.io/github/git-lfs/ubuntu noble InRelease
Ign:11 https://zotero.retorque.re/file/apt-package-archive ./ InRelease
Get:20 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:22 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Components [54.6 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,806 kB]
Ign:11 https://zotero.retorque.re/file/apt-package-archive ./ InRelease
Get:24 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
```

2.apt-get upgrade

This command is used to install the newest versions of all packages currently installed on the system. It uses the list generated by **apt-get update** to see which packages can be upgraded.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  network-manager-openvpn network-manager-openvpn-gnome
The following packages will be upgraded:
  brave-browser cloud-init docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin google
  libgphoto2-port12t64 libmtp-common libmtp-runtime libmtp9t64 libvirt-clients libvirt-daemon libvirt-dae
  libvirt-daemon-system libvirt-daemon-system-systemd libvirt-l10n libvirt0 libxnvctrl0 linux-base linux-
  software-properties-common software-properties-gtk systemd-hwe-hwdb wpasupplicant
34 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
Need to get 873 MB of archives.
After this operation, 32.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 https://brave-browser-apt-release.s3.brave.com stable/main amd64 brave-browser amd64 1.87.192 [130
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.3.0-1~ubuntu.2
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-hwe-hwdb all 255.1.7 [3,716 B]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libfprint-2-tod1 amd64 1:1.94.7+tod1-0ubu
```

3.apt-get install

This is the main command for installing new software packages. For example, running **sudo apt-get install vlc** would download and install the VLC media player.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get install vlc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
vlc is already the newest version (3.0.20-3build6).
0 upgraded, 0 newly installed, 0 to remove and 36 not upgraded.
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ █
```

4.apt-get remove

This command is used to uninstall a software package from the system. It removes the binary files but often leaves behind configuration files.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get remove firefox
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'firefox' is not installed, so not removed
0 upgraded, 0 newly installed, 0 to remove and 36 not upgraded.
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

5.apt-get clean

This command clears out the local repository of downloaded package files (.deb files). It removes everything from **/var/cache/apt/archives/** as well as its partial subdirectory, freeing up disk space.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get clean
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

6.apt-get autoclean

Similar to clean, but more selective. It only removes package files that can no longer be downloaded and are virtually useless. It clears out the .deb files from the cache for packages that are obsolete or have been superseded by newer versions.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get autoclean
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

7.apt-get autoremove

This command is used to remove packages that were automatically installed to satisfy dependencies for other packages but are no longer needed.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo apt-get autoremove
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 36 not upgraded.
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

8.Date

A simple command that displays the current system date and time. It shows the day, month, date, time, timezone, and year.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ date
Sun Mar  8 10:48:01 PM +06 2026
```

9.df

Short for "disk free", this command reports the amount of available disk space on all mounted file systems. It shows the total size, used space, available space, and the mount point for the drives.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ df
Filesystem            1K-blocks      Used Available Use% Mounted on
tmpfs                  786772        2620      784152   1% /run
/dev/nvme0n1p6 177757748 159383872 10333976  94% /
tmpfs                  3933848      432552   3501296  11% /dev/shm
tmpfs                  5120           8        5112   1% /run/lock
efivarfs               374          268        102  73% /sys/firmware/efi/efivars
tmpfs                  3933848         0   3933848   0% /run/qemu
/dev/nvme0n1p1   98304       51947    46357  53% /boot/efi
/dev/nvme0n1p2 312695732 169505068 127233468  58% /mnt/ExtSpace
tmpfs               786768        172    786596   1% /run/user/1000
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ du
```

10. du

Short for "disk usage", this command estimates and displays the file space usage of files or directories.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ du
4      .
```

11. free

This command displays the total amount of free and used physical memory (RAM) and swap space in the system. Adding the **-h** flag makes the output human-readable (showing values in MiB or GiB).

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ free
              total        used        free      shared  buff/cache   available
Mem:          7867696      3827220      2251028         714376       2851168       4040476
Swap:          4194300       2008356       2185944
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

12. uname -a

The `uname` command prints system information. The `-a` flag (`-a` for all) prints all available information, including the kernel name, hostname, kernel release, kernel version, machine hardware, and operating system.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ uname -a
Linux ahmed-0112230191 6.17.0-14-generic #14~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Jan 15 15:52:10 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

13. top

This is a real-time, dynamic view of a running system. It displays a list of processes currently being managed by the kernel, along with a summary of system resource usage (CPU, memory). It's like the Task Manager, but for Linux.

```

$ sudo adduser sheikh-tanvir
[sudo] password for tanvir-ahmed:
info: Adding user 'sheikh-tanvir' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'sheikh-tanvir' (1001) ...
top - 22:47:46 up 5:27, 1 user, load average: 1.29, 1.62, 3.11
Tasks: 333 total, 1 running, 331 sleeping, 0 stopped, 1 zombie
top - 22:52:13 up 5:31, 1 user, load average: 1.75, 1.77, 2.80
Tasks: 334 total, 2 running, 331 sleeping, 0 stopped, 1 zombie
%Cpu(s): 2.8 us, 1.2 sy, 0.0 ni, 95.7 id, 0.2 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7683.3 total, 2068.1 free, 3826.2 used, 2835.0 buff/cache
MiB Swap: 4096.0 total, 2136.3 free, 1959.7 used, 3857.1 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3329	tanvir-+	20	0	4942624	124456	42140	S	7.6	1.6	17:12.89	gnome-shell
4142	tanvir-+	20	0	48.9g	381596	178688	S	5.0	4.9	20:51.02	brave
4445	tanvir-+	20	0	49.1g	112504	107440	S	4.6	1.4	70:23.06	brave
2990	tanvir-+	20	0	25.0g	66160	43148	S	4.0	0.8	19:54.78	Xorg
4764	tanvir-+	20	0	1392.5g	532536	153172	S	1.7	6.8	5:46.44	brave
18090	tanvir-+	20	0	1392.5g	237112	110616	S	1.7	3.0	1:56.71	brave
651	root	-51	0	0	0	0	R	1.0	0.0	3:50.98	irq/145-rtw88_pci
908	root	20	0	407720	9972	9368	S	0.7	0.1	0:36.59	touchegg
19888	root	20	0	0	0	0	I	0.7	0.0	0:11.04	kworker/u32:4-events_power_efficient
80	root	-51	0	0	0	0	S	0.3	0.0	0:37.94	irq/9-acpi
907	root	20	0	424916	6616	6188	S	0.3	0.1	0:53.54	thermald
3367	tanvir-+	20	0	236060	6804	6484	S	0.3	0.1	0:02.91	at-spi2-registr
4186	tanvir-+	20	0	48.5g	78200	53972	S	0.3	1.0	5:20.94	brave
12996	root	0	-20	0	0	0	D	0.3	0.0	0:31.61	kworker/u33:4+i915_flip
13674	tanvir-+	20	0	1392.5g	335524	107356	S	0.3	4.3	3:56.23	brave
25885	root	20	0	0	0	0	I	0.3	0.0	0:05.21	kworker/u32:0-i915
25992	tanvir-+	20	0	1394.4g	196336	98976	S	0.3	2.5	0:26.52	brave
29139	tanvir-+	20	0	853296	147032	90440	S	0.3	1.9	0:08.83	flameshot
1	root	20	0	24132	11192	6908	S	0.0	0.1	0:03.80	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:01.52	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:36.59	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.37	migration/0

14. ps

Short for "process status", this command reports a snapshot of the current processes. Without any options, it shows only the processes running in the current shell. Using `ps aux` is a common way to show all running processes with detailed information.

```

sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ps
  PID TTY          TIME CMD
 27106 pts/0        00:00:00 bash
 29749 pts/0        00:00:00 ps

```

15. uptime

This very simple command tells how long the system has been running. It also shows the current time, how many users are logged on, and the system load averages for the past 1, 5, and 15 minutes.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ uptime
22:52:41 up 5:32, 1 user, load average: 1.57, 1.73, 2.75
```

16. ls

Short for "list", this command lists the contents of a directory. By default, it shows the files and folders in the current working directory. Common variations include `ls -l` (long format) and `ls -a` (show all, including hidden files).

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
a.txt b.txt
```

17. cd

Short for "change directory", this is one of the most fundamental commands. It allows to move between directories.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ mkdir sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ mkdir sample02
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
a.txt b.txt sample01 sample02
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ cd sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01/sample01$ pwd
/home/sheikh-tanvir/Assignment-01/sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01/sample01$
```

18. pwd

Short for "print working directory", this command outputs the full pathname of the directory that we are currently in. It's very useful for orienting oneself in the filesystem.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01/sample01$ pwd  
/home/sheikh-tanvir/Assignment-01/sample01  
sheikh-tanvir@ahmed-0112230191:~/Assignment-01/sample01$
```

19. rmdir

This stands for "remove directory". It is used to delete empty directories. If the directory contains any files, rmdir will not work and would need to use `rm -r` instead.

```
a.txt b.txt hi sample01 sample02  
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ rmdir sample01  
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls  
a.txt b.txt hi sample02  
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

20. mkdir

Short for "make directory", this command is used to create new, empty directories. For example, `mkdir NewFolder` will create a folder named "NewFolder" in the current location. We can also use `mkdir -p` to create parent directories as needed.


```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ touch a.txt b.txt
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
a.txt  b.txt
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ mkdir sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ mkdir sample02
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
a.txt  b.txt  sample01  sample02
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ cd sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01/sample01$ pwd
/home/sheikh-tanvir/Assignment-01/sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01/sample01$
```

21. cat

Short for "concatenate", this command has a few uses. Its most common use is to read and display the contents of a file directly in the terminal.

```
-bash: /usr/local/bin/nvim: Permission denied
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ sudo nvim a.txt
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ cat a.txt
Hi, from a.txt
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$
```

22. touch

This command is primarily used to update the timestamps (access and modification times) of a file. Although its most common use is to create a new, empty file if the filename provided does not already exist.

```
22.52.41 up 0.52, 1 user, load average: 1.57, 1.75, 2.75
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ touch a.txt b.txt
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
a.txt  b.txt
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ mkdir sample01
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ mkdir sample02
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ ls
a.txt  b.txt  sample01  sample02
```

23. history

This command displays a list of all the previously executed commands for the current user session.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ history
```

```
1 clear
2 ls
3 pwd
4 mkdir "Assignment-01"
5 ls
6 cd Assignment-01/
7 ls
8 clear
9 apt-get update
10 clear
11 sudo apt-get update
12 clear
13 apt-get update
14 clear
15 sudo apt-get update
16 clear
17 apt-get upgrade
18 clear
19 sudo apt-get upgrade
20 clear
21 sudo apt-get install vlc
22 sudo apt-get remove firefox
23 clear
24 sudo apt-get remove firefox
25 apt-get clean
26 clear
27 sudo apt-get clean
28 lear
29 clear
30 sudo apt-get clean
31 clear
32 sudo apt-get autoclean
33 clear
34 sudo apt-get autoremove
35 clear
36 date
37 df
38 du
39 free
40 uname -a
41 top
42 clear
```

24. whatis

This command is used to display a one-line manual page description for a given command. It gives a very brief summary of what a command does.

```
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ whatis whatis
whatis (1)          - display one-line manual page descriptions
sheikh-tanvir@ahmed-0112230191:~/Assignment-01$ █
```